Nicholas Wayne Geissler

908-309-3873 · nwg2009@nyu.edu · Courant Institute, NYU · nickgeissler.github.io

EDUCATION

PhD Mathematics, Courant Institute, New York University *2025 – present*

MSE & BS Applied Mathematics and Statistics, BS Computer Science, Johns Hopkins University · GPA 4.0/4.0 *2021 – 2025*

Coursework: Numerical Linear Algebra, Numerical PDE, Optimization, Machine Learning, Random Matrix Theory, Bayesian Statistics, Statistics, Probability Theory, Algorithms, Data Structures, Time Series Analysis

PUBLICATIONS & PREPRINTS

One-Step Generative Surrogate Models via Block-Triangular Joint Drifting
N. Geissler, S. Jha, R. Baptista, B. Peherstorfer *Submitted to SIAM Journal on Scientific Computing, 2026* · arXiv

First-Order Trajectory Matching: Fast Ensemble Predictions of Chaotic, Turbulent, Stochastic Systems
S. Jha, T. Schorlepp, **N. Geissler**, J. Berman, B. Peherstorfer *Accepted to NeurIPS 2026* · arXiv

Hankel Low-Rank Matrix Approximation for Gravitational-Wave Data Analysis
N. Geissler, V. Strokov, C. Kümmerle, S. Kushnarev, E. Berti *Accepted to Physical Review D, 2026* · arXiv

RESEARCH

Doctoral Researcher, Courant Institute, New York University *January 2026 – present*
Advisor: Dr. Benjamin Peherstorfer

- Adapted drifting schemes to build one-step generative surrogates for stochastic PDEs, sampling each time step in one network pass.
- Recast drifting to target the joint law of successive states, bypassing the lack of multiple samples from each conditional transition law.
- Benchmarked on GPUs against large diffusion and flow matching models, cutting energy error 4× and enstrophy error 3× relative to the best distilled baselines on 2D turbulence.

Undergraduate & Master's Researcher, Johns Hopkins University *January 2024 – May 2025*
Advisors: Dr. Sergey Kushnarev, Dr. Christian Kümmerle

- Denoised gravitational-wave time series via Hankel matrix embedding and structured low-rank approximation.
- Built a fast implementation of a low-rank denoising algorithm using FFT-based Hankel matrix products.
- Separated overlapping LISA-type wave sources with closely spaced frequencies, nearly matching optimal error scaling on synthetic benchmarks.
- Received a seed grant from the Space@Hopkins initiative.

INDUSTRY EXPERIENCE

Quantitative Trader Intern, TransMarket Group LLC *May – August 2024*
US Treasury Basis *Aguadilla, PR*

- Market-made US Treasury bonds and on-the-run bond futures.
- Developed an automated market-making algorithm for bond futures in Python; accelerated backtesting and parameter optimization through event-driven data subsampling.
- Proposed two trading strategies to desk based on statistical analysis; both are being further pursued.

Clinical Information Specialist, Merck & Co., Inc. *May 2023 – May 2025*
Quantitative Pharmacology and Pharmacometrics, Immune/Oncology *Rahway, NJ*

- Developed methods for statistical analysis of Phase 1 clinical trial data in R and MATLAB.
- Expanded existing algorithms to improve accuracy and account for covariates in two primary cancer indications.
- Leveraged novel evaluation techniques to isolate confounding variables in responder populations.

HONORS & AWARDS

- **Outstanding Master's Research:** one of the top master's theses in Applied Mathematics and Statistics at Johns Hopkins.
- **Applied Mathematics and Statistics Achievement Award:** outstanding accomplishment by a senior student at Johns Hopkins.
- **Naddor Prize:** significant achievement in departmental academic and extracurricular activities at Johns Hopkins.

TEACHING

Teaching Assistant, Linear Algebra, Courant Institute, New York University *September 2026 – present*
Teaching Assistant, Honors Statistics, Johns Hopkins University *August 2023 – May 2025*
Head Teaching Assistant, Honors Discrete Mathematics, Johns Hopkins University *January – May 2022*

PROGRAMMING

Software: Python, JAX, MATLAB, R, Java, C++, Git, Slurm